

- 6 (a) Define electric potential difference across a component.

[1]

- (b) A circuit contains four resistors and a battery of electromotive force (e.m.f.) 8.0 V with negligible internal resistance. When the variable resistor has resistance R , the currents in the circuit are 0.030 A , I_1 and I_2 , as shown in Fig. 6.1.

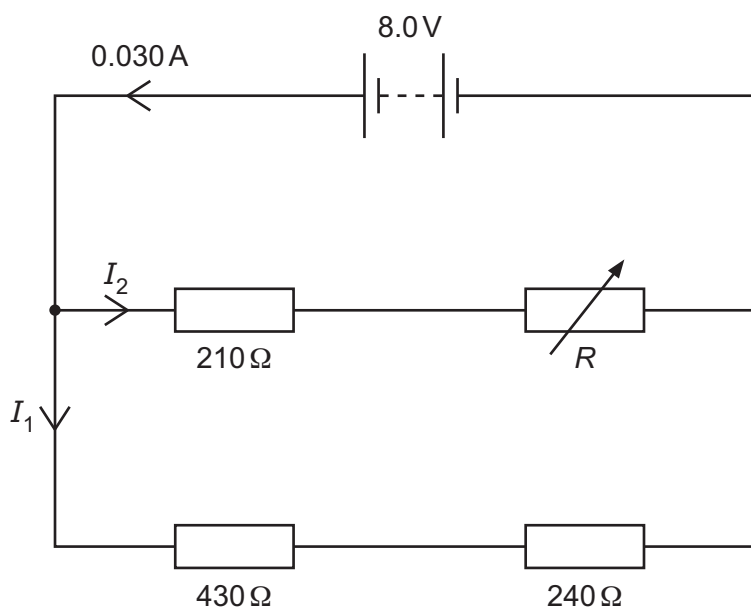


Fig. 6.1

- (i) Determine the charge passing through the battery in a time of 4.0 minutes.

[2]

- (ii) Calculate I_1 .

[2]

- (iii) Calculate I_2 .

[1]

- (iv) Determine R .

[2]

- (c) The variable resistor in (b) is fitted with a scale so that its resistance can be accurately determined.

The resistor of resistance $240\ \Omega$ is now replaced by a new resistor X of unknown resistance. A galvanometer is connected as shown in Fig. 6.2.

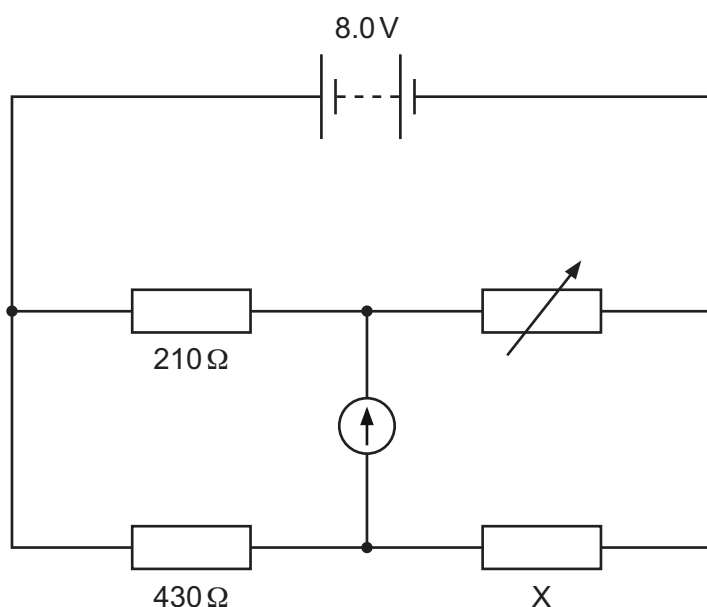


Fig. 6.2

With reference to ratios of resistances, explain how this circuit can be used to determine the resistance of X .

[2]

[Total: 10]